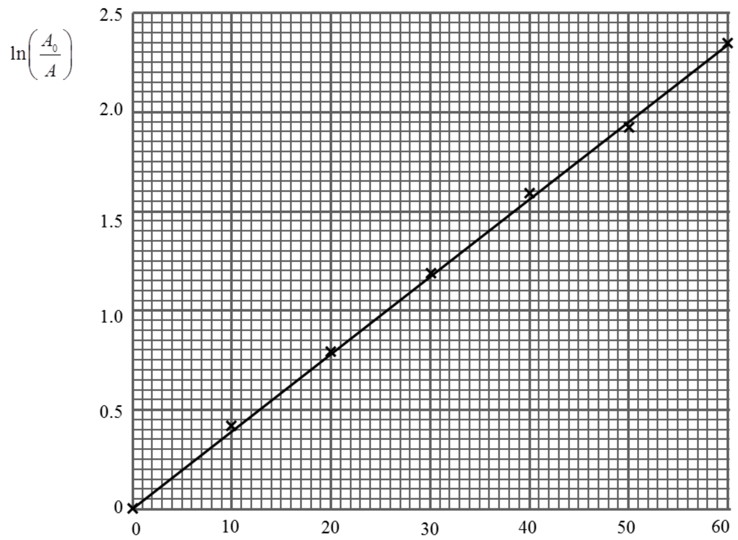
Question			Marking details	AO1	AO2	AO3	Total	Maths	Prac
				Marks available					
6	*(a)*	(i)	$kx = mg$ (1) $x = \frac{0.150 \times 9.81}{7.5}$ = 0.196 m answer (1)		2		2	2	2
		(ii)	$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.150}{7.5}}$ substitution (1) = 0.889 s (1) NB. No credit for use of $T = 2\pi\sqrt{\frac{l}{g}}$	1	1		2	2	2
	(b)		Drop in amplitude at beginning and end compared, e.g. 0.033 m and 0.005 m over first and last intervals. [Or rates of decrease: 0.0033 m per oscillation and 0.0005 m per oscillation] consider rate of decrease (1)] Justify by noting the magnitude is larger at the start than at end of experiment. (1)			2	2		2
	(c)	(i)	When n = 0, $A = A_0 e^{-\frac{0}{N}} = A_0 e^0 = A_0$ [Alternative: say $e^0 = 1$, so $A = A_0$.]			1	1	1	1
		(ii)	$A = A_0 e^{-\frac{n}{N}}$; $0.029 = 0.095 e^{-\frac{30}{N}}$ substitution (1) $N = \frac{-30}{\ln\left(\frac{0.029}{0.095}\right)}$ or equiv → N =25.28 ≅ 25 answer (1)	1	1		2	2	2
	(d)	(i)	(table below) All three values correct (1) [accept 1.6, 1.9, 2.4 if consistent]		1		1	1	1

Oscillation number (n)	Amplitude (A) /m	
0	0.095	0
10	0.062	0.43
20	0.043	0.79
30	0.029	1.19
40	0.019	1.61
50	0.014	1.91
60	0.009	2.36

		(ii)	 Labelled axes with a suitable choice of scales (1) All points plotted correctly including origin (1) **(ecf)** Appropriate line of best fit [not necessarily through origin] ecf (1)	1	1 1		3	3	3
		(iii)	Any two of: 1) It is a straight line, or positive gradient (1) 2) Passes through the origin (1) 3) The measured points lie essentially on a straight line or little scatter(1)			2	2		2
		(iv)	Gradient = $\frac{2.35-0}{60-0}$ (from graph) = 0.039 **ecf** (1) $N = \frac{1}{\text{gradient}}$ (1) = 25.6 (1) Accept answers in range 25 – 26. Use of a point on the line – allow. Use of a data point not on the line $\rightarrow$ 1st mark not available.			3	3	3	3
		(v)	Expected to be the one from *(d)*(iv) as straight line based on all points [The value in *(c)*(ii) is calculated from only a single point, which may be anomalous.]			1	1		1
			Question 6 total	**3**	**7**	**9**	**19**	**14**	**19**